ESP32 SMART HOME

System Design and Demonstration

Project Team

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Presentation Outline

01 Introduction

02 Requirements

03 System Design

04 Budget

05 Progress and Demo

06 Team and Status

1. Problem and Users

The problem

In most homes, turning on a light still means walking to the switch. Even with a smart app, you still have to press the button yourself — there is no way to tell the system when the light should turn on, so it can never act automatically.

Intended users

Homeowners and renters who want everyday devices controllable from a phone, with no technical knowledge required.

Why it matters

- Control devices without going to each switch.
- See device state live, with history for monitoring.
- Automate device control through customizable rules.

1. Checkpoint 1 Scope

In scope now

- Light ON/OFF and door open/close over MQTT (`smarthome/cmd`) from the web dashboard.
- PIR motion turns the light on, auto-off after inactivity.
- RFID allowlist opens the door; unknown cards are rejected and logged.
- Live web dashboard: SSE stream of the last 300 events, device state at a glance.
- Rules engine: trigger/condition/action rules, persisted in `rules.json` across restarts.

Deliberately deferred

- Mobile app on the unified contract — alignment with the refactored topics is in progress.
- Voice control (planned after core acceptance).
- Database-backed event history; logs stay in a RAM buffer (only rules persist).
- Cloud/remote access; the demo runs on a shared LAN.
- Energy-saving measurement.

2. Functional Requirements

ID	Required behavior
FR1	The phone turns the light ON and OFF.
FR2	The phone opens and closes the model door.
FR3	PIR detection turns the light on, followed by off after a configurable inactivity period.
FR4	RFID checks an allowlist and opens the door for an authorized card.
FR5	The app displays light, door, motion and connection state.
FR6	The app shows recent sensor, access and device events.

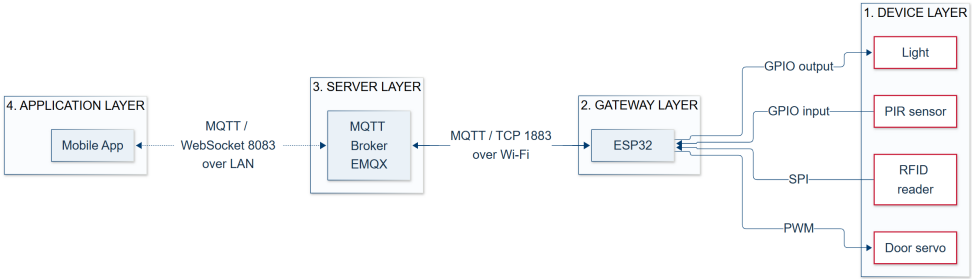
These are acceptance requirements from the task sheet. Voice is planned after core acceptance.

2. Non-functional Targets

ID	Quality requirement	Acceptance criterion
NFR1	Operational stability	Run for 30 minutes without unintended reset or connection loss.
NFR2	Mechanical reliability	Complete 20 door cycles without jamming.
NFR3	Recovery and consistency	Reconnect after a network/broker interruption and restore current device states.
NFR4	Input robustness	Reject and log invalid control payloads while keeping firmware running.
NFR5	Reproducibility	All Must acceptance scenarios pass twice consecutively.

Targets come from M3-11, M3-08, M1-03/04/10, M1-06 and M5-09. Measurement results are pending.

3. Four-layer Architecture



The original components form four logical layers. ESP32 acts as the gateway. The mobile connection needs MQTT contract alignment and WebSocket configuration.

3. MQTT Contract per Hop

	Hardware / test server	Current mobile code
Command topic	`smarthome/cmd`	`smarthome/cmd/light` `smarthome/cmd/door`
Feedback topics	`smarthome/logs`	`smarthome/state/+` `smarthome/event/access`
Payload	JSON command and log schema	Text commands and state values, JSON access events
Broker transport	MQTT over TCP 1883	MQTT over WebSocket 8083
Integration priority	Agree one version, update both clients and rerun shared test cases.	

3. Technology Choices and Rationale

Technology	Rationale for this prototype
ESP32 / PlatformIO	Wi-Fi and device interfaces in a shared firmware project.
MQTT / EMQX	Pub/sub messaging for commands, states and events.
React Native / Expo	Shared mobile screens with MQTT connectivity.
Python + Paho	Test commands and display incoming device logs.
Docker Compose	Repeatable broker and service setup on a laptop.
Log storage	Server RAM holds 300 logs. App storage retains 50 logs and settings.

Core scope uses session/local logs. Persistent server storage remains a future design decision.

4. Bill of Materials

Component / set	Availability / budget (VND)
ESP32 WT-01 (Ethernet and Wi-Fi)	Available
USB–TTL CH340 adapter	Available
MB-102 breadboard	Available
SG90 servo	46,116
RC522 RFID module	Available
5V relay	Available
HC-SR501 PIR sensor	34,560
LM2596 buck module	45,144
Jumper wires (three types)	Available
12V 2A adapter	46,116
DC power jack	11,124
Budget for remaining items	**183,060**

Available items are already owned. RC522 selection remains to be confirmed.

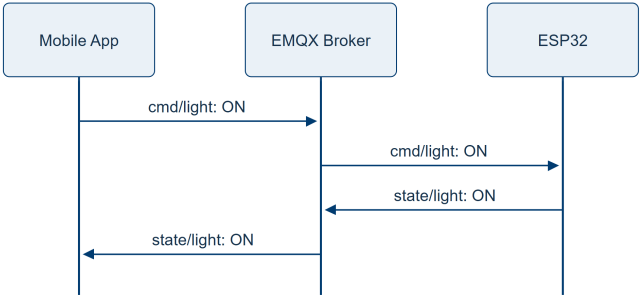
5. End-to-end Data Flow



The hardware branch contains this path in code. The server stores recent logs in RAM and serves the web dashboard.

Durable database storage is a pending scope decision. Physical sensor-to-screen evidence still needs to be recorded.

5. Light Command and State Feedback



Original contract with topic prefix `smarthome/`. Alignment with the current JSON schema is pending.

Hardware Demo

Add the model photo, wiring view or live demonstration here.

App Demo

Add phone screenshots, a recording or the live app here.

6. Team, Repository and Status

Member	Name	Student ID	Primary ownership
M1	Trần Sỹ Nguyên	20235985	Wi-Fi/MQTT core and firmware interfaces
M2	Lăng Hồng Nguyệt Anh	20235885	PIR, RFID authorization and automation
M3	Lương Ngọc Vũ Long	20235967	Power, wiring, actuators and physical model
M4	Hà Đức Dương	20235923	Mobile dashboard, MQTT control and event UI
M5	Nguyễn Vũ Anh Khoa	20235957	Broker, integration, QA and report coordination

github.com/itstheblank/esp32_smart_home

Shared project sheet

Thank You!

Questions and Discussion